

Question 1

- **Positivity:** We can always cover E with itself, and choose some $i \in I$ and let $J = \{i\}$ and $E_i = E$, then $E = \bigcup_{j \in J} E_j$, and thus $\mu(E) \geq \mu_i(E) \geq 0$ since μ_i is a positive measure.
- **Empty Set:** Clearly, as in Positivity, this supremum set is not empty so suppose $(E_i)_{i \in J} \subset \mathcal{E}$ is pairwise disjoint and $\emptyset = \bigcup_{i \in J} E_i$ and $J \subset I$ is at most countable. Since $\emptyset = \bigcup_{i \in J} E_i$, we must have $E_i = \emptyset$ for all $i \in J$. So

$$\sum_{i \in J} \mu_i(E_i) = \sum_{i \in J} \underbrace{\mu_i(\emptyset)}_{=0} = 0$$

since each μ_i is a positive measure. Thus $\mu(\emptyset) = 0$

- **σ -additivity:** Suppose that $(E_n)_{n \in \mathbb{N}} \subset \mathcal{E}$ is pairwise disjoint. Let $\varepsilon > 0$. Then, for each $n \in \mathbb{N}$, we have a family $(E_{nm})_{m \in J_n} \subset \mathcal{E}$ that is pairwise disjoint and $E_n = \bigcup_{m \in J_n} E_{nm}$ and $J_n \subset I$ is at most countable, and

$$\mu(E_n) - \varepsilon 2^{-n} < \sum_{m \in J_n} (\mu_m(E_{nm}))$$

so

$$\sum_{n \in \mathbb{N}} (\mu(E_n)) - \varepsilon < \sum_{n \in \mathbb{N}} \sum_{m \in J_n} (\mu_m(E_{nm}))$$

Since the summands are nonnegative, we can use Tonelli's Theorem for sums and swap indices. To do so, let $J = \bigcup_{n \in \mathbb{N}} J_n$, $N_j = \{n \in \mathbb{N} : j \in J_n\}$ for $j \in J$ and $A_j := \bigcup_{n \in N_j} E_{nj}$. J is at most countable, as a countable union of at most countable sets. Since $(E_{nm})_{n \in \mathbb{N}, m \in J_n}$ is pairwise disjoint, so is $(A_j)_{j \in J}$. Moreover,

$$\bigcup_{n \in \mathbb{N}} E_n = \bigcup_{n \in \mathbb{N}, j \in J_n} E_{nj} = \bigcup_{j \in J} A_j$$

$$\text{So } \mu\left(\bigcup_{n \in \mathbb{N}} E_n\right) \geq \sum_{j \in J} \mu_j(A_j) = \sum_{j \in J} \mu_j\left(\bigcup_{n \in N_j} E_{nj}\right) = \sum_{j \in J} \sum_{n \in N_j} \mu_j(E_{nj}) = \sum_{n \in \mathbb{N}} \sum_{j \in J_n} \mu_j(E_{nj}) > \sum_{n \in \mathbb{N}} (\mu(E_n)) - \varepsilon$$

This is true for all $\varepsilon > 0$ and thus

$$\mu\left(\bigcup_{n \in \mathbb{N}} E_n\right) \geq \sum_{n \in \mathbb{N}} (\mu(E_n))$$

To show the other inequality, let $E = \bigcup_{n \in \mathbb{N}} E_n = \bigcup_{j \in J} B_j$ for $(B_j)_{j \in J} \subset \mathcal{E}$ pairwise disjoint and $J \subset I$ is at most countable. Now, noticing that $(B_j \cap E_n)_{j \in J}$ is p.d. and

$$\sum_{j \in J} \mu_j(B_j) = \sum_{j \in J} \mu_j\left(\bigcup_{n \in \mathbb{N}} B_j \cap E_n\right) = \sum_{j \in J} \sum_{n \in \mathbb{N}} \mu_j(B_j \cap E_n) = \sum_{n \in \mathbb{N}} \sum_{j \in J} \mu_j(B_j \cap E_n)$$

Then, since $E_n = \bigcup_{j \in J} (B_j \cap E_n)$, with $(B_j \cap E_n)_{j \in J}$ being pairwise disjoint, we get

$$\sum_{j \in J} \mu_j(B_j \cap E_n) \leq \mu(E_n)$$

So finally,

$$\sum_{j \in J} \mu_j(B_j) \leq \sum_{n \in \mathbb{N}} \mu(E_n)$$

meaning

$$\mu \left(\bigcup_{n \in \mathbb{N}} E_n \right) \leq \sum_{n \in \mathbb{N}} (\mu(E_n))$$

making μ σ -additive and thus a positive measure

Question 2

- (a) • **Empty set:** For any set $A \in \mathcal{B}(X)$, clearly $\mu(A) \geq 0$ so $\Phi(E) \geq 0$ for all $E \subset X$ (note that X itself is open, so $\{\mu(A) : A \text{ is open, } A \supset E\}$ is nonempty). On the other hand, $\emptyset$ is open and $\mu(\emptyset) = 0$, so $\Phi(\emptyset) = 0$
- **σ -subadditivity:** Let $A \subset X$, $(A_n)_{n \in \mathbb{N}} \subset \mathcal{P}(X)$ such that $A \subset \bigcup_{n \in \mathbb{N}} A_n$. Let $\varepsilon > 0$. For each $n \in \mathbb{N}$, there exists an open set $O_n \supset A_n$ such that, since Φ is an infimum, and that μ is finite,

$$\Phi(A_n) + \varepsilon 2^{-n} > \mu(O_n)$$

Then $\bigcup_{n \in \mathbb{N}} O_n \supset \bigcup_{n \in \mathbb{N}} A_n \supset A$ and $\bigcup_{n \in \mathbb{N}} O_n$ is open. So, since μ is σ -subadditive,

$$\Phi(A) \leq \mu \left(\bigcup_{n \in \mathbb{N}} O_n \right) \leq \sum_{n \in \mathbb{N}} \mu(O_n) \leq \sum_{n \in \mathbb{N}} (\Phi(A_n) + \varepsilon 2^{-n}) = \Phi(A) + \varepsilon$$

This is true for all $\varepsilon > 0$ and so

$$\Phi(A) \leq \sum_{n \in \mathbb{N}} \Phi(A_n)$$

- (b) Let $E_1, E_2 \subset X$ with $d(E_1, E_2) = r > 0$. From σ -subadditivity, $\Phi(E_1 \cup E_2) \leq \Phi(E_1) + \Phi(E_2)$. Now $d(E_1, E_2) = \inf_{(e_1, e_2) \in E_1 \times E_2} d(e_1, e_2)$. For each $e_1 \in E_1$, clearly $B(e_1; \frac{1}{2}r) \cap E_2 = \emptyset$. Consider the open sets $U_1 = \bigcup_{e_1 \in E_1} B(e_1; \frac{1}{2}r)$ and $U_2 = \bigcup_{e_2 \in E_2} B(e_2; \frac{1}{2}r)$. Then $E_1 \cup E_2 \subset U_1 \cup U_2$ and $U_1 \cap U_2 = \emptyset$ (as if $x \in U_1 \cap U_2$, then there exist some $e_1 \in E_1, e_2 \in E_2$ such that $d(e_1, x) < \frac{1}{2}r$ and $d(e_2, x) < \frac{1}{2}r$, then $d(e_1, e_2) < r$, a contradiction). Let $O \supset E_1 \cup E_2$ be open. Then $O \cap U_1 \supset E_1$ and $O \cap U_2 \supset E_2$ and both are open, so, using the additivity of the measure μ ,

$$\mu(O) \geq \mu(O \cap (U_1 \cup U_2)) = \mu((O \cap U_1) \cup (O \cap U_2)) = \mu(O \cap U_1) + \mu(O \cap U_2) \geq \Phi(E_1) + \Phi(E_2)$$

So $\Phi(E_1 \cup E_2) \geq \Phi(E_1) + \Phi(E_2)$ and

$$\Phi(E_1) + \Phi(E_2) = \Phi(E_1 \cup E_2)$$

Question 3

($\Rightarrow$) Suppose f is measurable. f^2 is measurable since $f^2 = f \cdot f$, as the product of measurable functions, and $\{x \in X : f(x) > 0\} = f^{-1}((0, \infty)) \in \mathcal{E}$ since $(0, \infty)$ is a Borel set.

($\Leftarrow$) Let $t \in \mathbb{R}$. We need to show that $\{x \in X : f(x) > t\} \in \mathcal{E}$. The case $t = 0$ is given.

If $t > 0$, then we have that $f(x) > t \iff f(x) > 0$ and $|f(x)| > |t| \iff f(x) > 0$ and $f(x)^2 > t^2$, giving $\{x \in X : f(x) > t\} = \{x \in X : f(x) > 0\} \cap \{x \in X : f(x)^2 > t^2\} \in \mathcal{E}$.

If $t < 0$, then we have that $f(x) > t \iff f(x) > 0$ or $(f(x) \leq 0 \text{ and } |f(x)| < |t|) \iff f(x) > 0$ or $(f(x) \leq 0 \text{ and } f(x)^2 < t^2)$, giving

$$\{x \in X : f(x) > t\} = \{x \in X : f(x) > 0\} \cup (\{x \in X : f(x) \leq 0\} \cap \{x \in X : f(x)^2 < t^2\}) \in \mathcal{E}$$

So for all $t \in \mathbb{R}$, $f^{-1}((t, \infty)) \in \mathcal{E}$ making f measurable